

Prediction for House Rent Price in Mogadishu Districts

1. Title & Collection Method

Title

Prediction for House Rent Price in Mogadishu Districts

Collection Method

This dataset focuses on predicting monthly house rental prices across different districts in Mogadishu, Somalia. The dataset was collected by compiling rental housing information from various property listings and observations. Each record represents a single house and contains information about its location, type, number of rooms, security availability, parking availability, and rental price.

The dataset contains **100 samples (rows)** and is designed to analyze how housing characteristics influence rental prices.

2. Description of Features & Labels

This dataset contains information related to residential rental properties in Mogadishu.

Features (X)

- ✓ **District_Name** – The district where the house is located (e.g., Hodan, Wadajir, Yaqshid).
- ✓ **House_Type** – Type of property such as Apartment, Villa, or Jingad.
- ✓ **Rooms** – Number of rooms available in the house.
- ✓ **Security** – Indicates whether security services are available (Yes/No).
- ✓ **Parking** – Indicates whether parking space is available (Yes/No).

Target Label (y)

- ✓ **Price** – Monthly rental price of the house in US Dollars (USD).

The machine learning model will use the five input features to predict the rental price.

3. Dataset Structure

The dataset consists of **100 rows (samples)** and **6 columns (variables)**.

Sample Table

District_Name	House_Type	Rooms	Security	Parking	Price (USD)
Hodan	Apartment	3	Yes	Yes	450
Wadajir	Villa	5	Yes	Yes	850
Yaqshid	Apartment	2	No	No	300
Heliwa	Jingad	4	Yes	No	500
Karaan	Apartment	3	No	Yes	400

The table above illustrates the structure of the dataset where each row represents a rental property and each column represents a specific attribute.

4. Quality Issues

A preliminary examination of the dataset reveals several potential data-quality issues that should be addressed during preprocessing.

a. Missing Values

Some records may contain missing information in fields such as:

- Security
- Parking
- Rooms
- Price

Missing values can reduce model accuracy and should be handled using imputation or record removal.

b. Typographical Errors

District names or house types may contain spelling inconsistencies such as:

- "Wadajir" vs "Wadajiir"
- "Apartment" vs "apartment"

These inconsistencies create duplicate categories and must be standardized.

c. Duplicate Records

The same house listing may appear multiple times in the dataset, leading to duplicate observations.

d. Categorical Imbalance

Certain districts or house types may appear more frequently than others. For example, apartments may significantly outnumber villas.

This imbalance may affect model performance.

e. Data Formatting Issues

Categorical variables such as Security and Parking must be converted into numerical values before model training.

Examples:

- Yes $\rightarrow$ 1
- No $\rightarrow$ 0

5. Learning Type

Supervised Learning

This problem is classified as a **Supervised Learning** problem.

Justification

Supervised learning requires both:

- Input Features (X)
- Output Label (y)

In this dataset, the target variable is clearly defined as:

Price (USD)

Because each house record contains both the property characteristics and its corresponding rental price, the algorithm can learn the relationship between them and make future predictions.

Therefore, this dataset belongs to **Supervised Machine Learning**.

6. Use Case

a. Machine Learning Applications

i. Regression

The primary use of this dataset is **Regression** because the target variable (Price) is continuous.

Examples of suitable algorithms include:

- Linear Regression
- Decision Tree Regression
- Random Forest Regression

These models can estimate rental prices for new houses based on their characteristics.

ii. Classification

The dataset can also be transformed into a classification problem by grouping rental prices into categories such as:

- Low Rent
- Medium Rent
- High Rent

A classification model can then predict which category a house belongs to.

iii. Clustering

If the Price column is removed, clustering algorithms such as K-Means can identify groups of similar houses based on:

- District
- House Type
- Rooms
- Security
- Parking

This can help real-estate agencies identify market segments.

b. Placement in the Data Science Lifecycle

i. Problem Definition

The goal is to predict house rental prices in Mogadishu districts.

ii. Data Collection

Housing information is gathered and stored in a structured dataset.

iii. Data Preparation

The dataset is cleaned by:

- Handling missing values
- Removing duplicates
- Correcting spelling errors
- Encoding categorical variables

iv. Modeling

Machine learning algorithms are trained using historical rental data.

v. Evaluation

Model performance is measured using metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

vi. Deployment

The final model can be integrated into a rental price prediction system used by landlords, tenants, and real-estate companies.

Conclusion

The "Prediction for House Rent Price in Mogadishu Districts" dataset provides a practical example of a supervised machine learning problem. By using housing characteristics such as district, house type, number of rooms, security, and parking facilities, machine learning models can accurately estimate rental prices. After preprocessing and cleaning, the dataset can support regression, classification, and clustering applications while contributing to data-driven decision-making in the real-estate sector.